



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Self-Calibrating Question Bank Engine Using Real-Time Learner Response Analytics

CSA0925 – Programming in Java

Guided By:

Dr. C. Sivasankar

Presented By:

D. Vahida (192472268)

J. Renuka Devi (192573007)

PROBLEM BACKGROUND

Application Domain

Education Technology (EdTech), Online Assessment Systems, and Adaptive Learning Platforms that aim to improve the quality and fairness of student evaluations.

Current Scenario

- Question difficulty is assigned manually and remains static over time.
- Assessments are not personalized due to limited learner performance analysis.

Need for the Project

- Automatically update question difficulty using real-time learner responses.
- Generate personalized assessments for improved learning and evaluation.

Target Users

Students, Faculty, Educational Institutions, Online Learning Platforms



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PROBLEM STATEMENT

Problem Definition

Current question banks use fixed difficulty levels, making assessments less adaptive and less effective for diverse learners.

Limitations

- Difficulty is updated manually.
- No real-time learner response analysis.
- Same questions for different skill levels.

Expected Impact

- Dynamic difficulty calibration.
- Personalized question recommendations.
- Improved assessment quality.

OBJECTIVES

Primary Objective

Develop an intelligent question bank that automatically updates question difficulty using real-time learner responses.

Measurable Objectives

- Collect learner responses in real time.
- Calculate question difficulty automatically.
- Recommend personalized practice questions.
- Generate balanced examinations.
- Provide instructor analytics dashboard.

Project Scope

- ❖ Web-based application
- ❖ Java backend
- ❖ Database integration
- ❖ Adaptive question recommendation

LITERATURE SURVEY

Paper	Method Used	Key Contribution	Research Gap
Dai et al. (2023)	Adaptive Testing + Recommendation	Used adaptive testing techniques for personalized learning recommendations.	Limited continuous question difficulty updates.
Liu et al. (2024)	Machine Learning + Computerized Adaptive Testing	Reviewed ML-based approaches for adaptive question selection.	Requires better real-time learner response integration.
Wang et al. (2024)	Deep Reinforcement Learning	Proposed intelligent question selection strategies for adaptive tests.	Focuses mainly on selection, not complete question bank calibration.
Sharpnack et al. (2024)	AutoML + Item Response Theory	Improved automatic item calibration using ML techniques.	Does not include personalized practice recommendation.



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EXISTING VS PROPOSED SYSTEM

Existing System	Proposed System
Manual difficulty	Automatic calibration
Static question bank	Dynamic question bank
Same test for everyone	Personalized tests
No analytics	Real-time analytics
Limited recommendations	Smart recommendations

Advantages:

- Adaptive learning
- Faster evaluation
- Improved engagement
- Reduced manual effort

Novelty

Uses continuous learner response analytics to update question difficulty dynamically.



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REQUIREMENTS

Functional Requirements

- Capture learner responses & performance data.
- Analyze question difficulty dynamically
- Generate adaptive tests
- Provide personalized recommendations

Non-Functional Requirements

- High accuracy
- Scalability for multiple users
- Secure data handling
- Fast processing

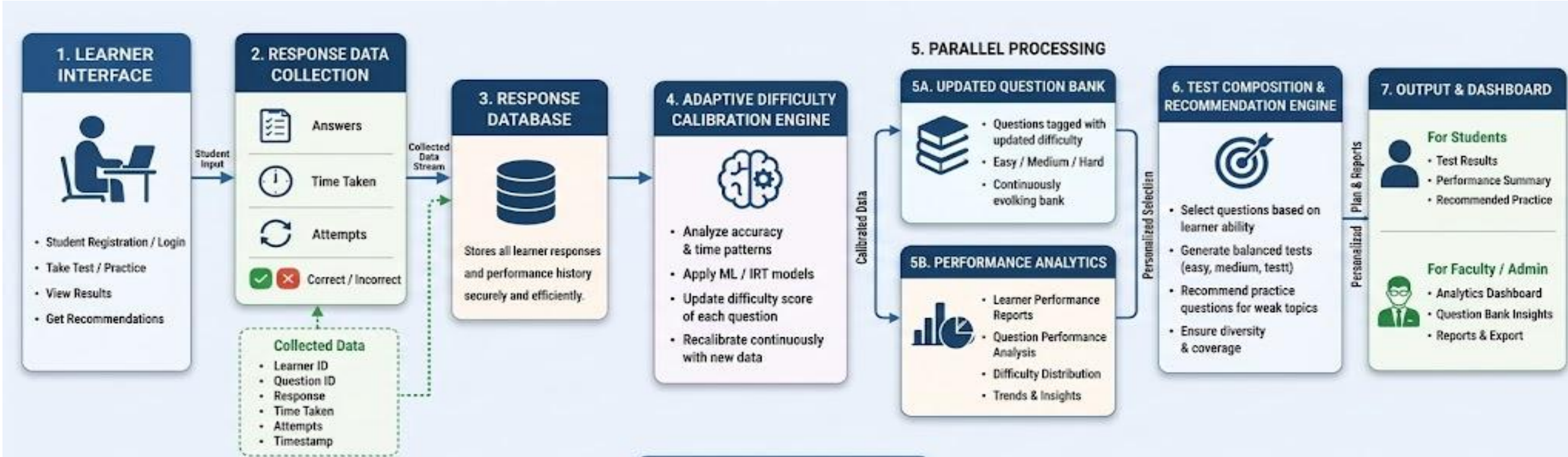
Software Requirements

- Java
- Database (MySQL)
- Web Framework

Dataset / API

- Learner response data
- Accuracy, time, attempts, performance history

SYSTEM ARCHITECTURE



MODULE 1: RESPONSE DATA COLLECTION

To Collect Real time student data fro the question engine. Includes gathering question details.

MODULE 2 : ADAPTIVE DIFFICULTY CALIBRATION

Dynamically adjust question difficulty based on student performance.

MODULE 3: TEST COMPOSITION & RECOMMENDATION

Create adaptive assessments tailored to individual learning progress.



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PLANNING



Project Execution Plan

Phase	Planned Activities
Phase 1: Research	Literature survey, problem analysis, requirement identification
Phase 2: Design	System architecture, module design, technology selection
Phase 3: Development	Dataset preparation, response analytics, difficulty calibration model
Phase 4: Integration	Adaptive test generation and recommendation system
Phase 5: Evaluation	Testing, performance analysis, and documentation

EXPECTED OUTCOMES

Conclusion

- Develops an intelligent question bank that adapts to learner performance.
- Uses real-time analytics to update question difficulty automatically.
- Enables personalized and effective assessments.

Benefits

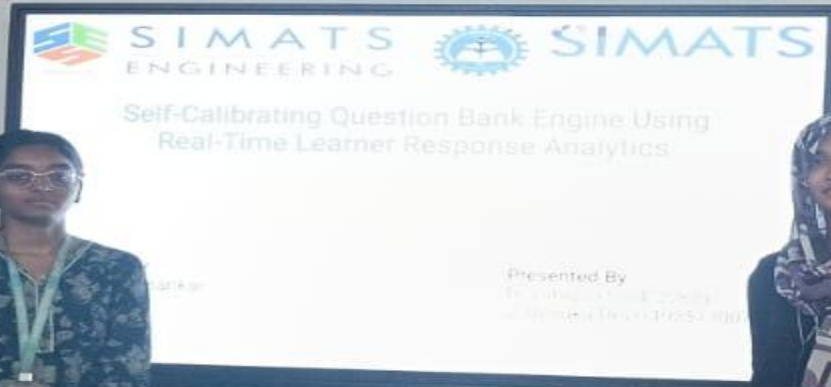
- Improves learning experience through adaptive practice.
- Provides accurate evaluation of learner ability.
- Reduces manual effort in managing question difficulty.

Future Scope

- Integration with LMS and online learning platforms.
- Advanced AI models for better personalization.
- Real-time instructor analytics dashboard.

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